

ASSIGNMENT-3

SUB:TOC

1) Define the following with suitable examples: (i) Ambiguous Language (ii) Chomsky Normal Forms (iii) Greibach Normal Form

2) Convert the following grammar to Chomsky Normal form. $S \rightarrow A|AB0|A1A$ $A \rightarrow A0|\epsilon$
 $B \rightarrow B1|BC$ $C \rightarrow CB|CA|1B$.

3) Prove that the language $L = \{a^n b^n c^n | n \geq 1\}$ is not context free using pumping lemma.

4) Eliminate the unit production of the following grammar

$S \rightarrow A|bb$ $A \rightarrow B|b$ $B \rightarrow S|a$

6) Remove all the ϵ -productions and Unit productions from the grammar: $S \rightarrow aA | aBB$ $A \rightarrow aAA | \epsilon$
 $B \rightarrow bB | bbC$ $C \rightarrow B$

7) Obtain a grammar in Chomsky normal form (CNF) equivalent to grammar G with the production P given $S \rightarrow aAbB$ $A \rightarrow Aa|a$ $B \rightarrow Bbb|b$

8) Define GNF. Convert the following grammar into GNF. $S \rightarrow AB1 | 0$ $A \rightarrow 00A | B$ $B \rightarrow 1A1$

9) Obtain the following grammar in CNF: $S \rightarrow ASB | \epsilon$ $A \rightarrow aAS | a$ $B \rightarrow SbS | A | bb$

10) Define Turing Machine. With a neat Block diagram, explain the the working of basic Turing Machine.

11) Design a Turing Machine to accept all set of palindrome over $\{a,b\}^*$. Draw the transition table and also transition diagram. Show the sequence of IDs for the string: "ababa"

12) Design a Turing Machine to accept all set multiplication . Draw the transition table and also transition diagram.

13) Design a Turing Machine to accept all set of symbol of language $L = \{a^n b^n | n \geq 1\}$ over $\{a,b\}^*$. Draw the transition table and also transition diagram. Show the sequence of IDs for the string: "aaabbb"

14) Design a Turing Machine to accept strings formed on $\{0,1\}^*$ and ending with 000. Write transition diagram and sequence of IDs for $w = 101000$

15) Write a short note on: a) Multi tape Turing Machine b) Nondeterministic Turing Machine